FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VI

JEE (Main)-2021

TEST DATE: 14-02-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Velocity of a particle moving in a curvilinear path in a horizontal X Y plane varies with time as $\vec{v}=\left(2t\,\hat{i}+t^2\,\hat{j}\right)m/s$. Here, t is in second. At $t=1s$.

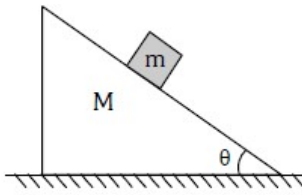
   (A) Acceleration of particle is $8\,m/s^2$

   (B) Tangential acceleration of particle is $\frac{4}{\sqrt{5}}m/s^2$

   (C) Radial acceleration of particle is $\frac{6}{\sqrt{5}}m/s^2$

   (D) Radius of curvature to the path is $\frac{5\sqrt{5}}{2}m$

2. A block of mass m slides down on a wedge of mass M as shown in figure. Let $\vec{a}_1$ be the acceleration of the wedge and $\vec{a}_2$ the acceleration of block w.r.t. ground. N_1 is the normal reaction between block and wedge and N_2 the normal reaction between wedge and ground. Friction is absent everywhere. Select the incorrect alternative



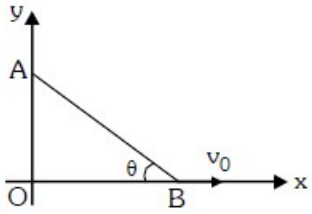
   (A) $N_2<(M+m)g$

   (B) $N_1=m\left(g\cos\theta-\left|\vec{a}_1\right|\sin\theta\right)$

   (C) $N_1\sin\theta=M\left|\vec{a}_1\right|$

   (D) $m\vec{a}_2=-M\vec{a}_1$

3. The end B of the rod AB which makes angle θ with the floor is being pulled with, a constant velocity v_0 as shown. The length of the rod is ℓ. At the instant when $\theta=37°$ then



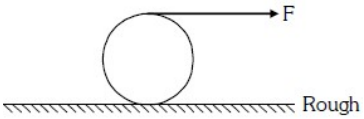
   (A) Velocity of end A is $\frac{5}{3}v_0$ downwards

   (B) Angular velocity of rod is $\frac{5}{3}\frac{v_0}{\ell}$

   (C) Angular velocity of rod is constant

   (D) Velocity of end A is constant

4. A solid sphere of radius R is rolled by a force F acting at the top of the sphere as shown in the figure. The sphere rolls without slipping on rough stationary surface. Initially sphere is in the rest position then:



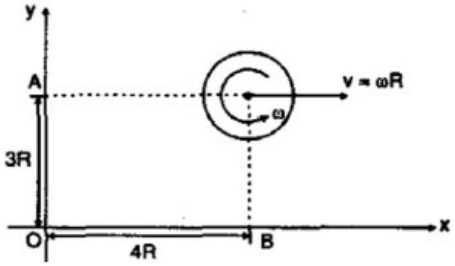
(A) Work done by friction when the centre of mass moves a distance S is $2FS$.

(B) Speed of the CM when CM moves a distance S is $\sqrt{\frac{20}{7}\frac{FS}{M}}$

(C) Work done by the force F when Centre of Mass moves a distance S is FS

(D) Speed of the CM when CM moves a distance S is $\sqrt{\frac{4RS}{M}}$

5. A disc of mass M and radius R moves in the x-y plane as shown in the figure. The angular momentum of the disc at the instant shown is:



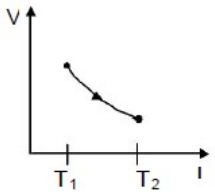
(A) $\frac{5}{2}mR^2\omega$ along -z axis about O

(B) $\frac{5}{2}mR^2\omega$ along $+z$ axis about O

(C) Zero about point A

(D) $4mR^2\omega$ along $+Z$ axis about A

6. The figure shows a process AB undergone by 2 moles of an ideal diatomic gas. The process AB is in such a way that VT=constant. T_1=300K and T_2=500K (R= gas constant)



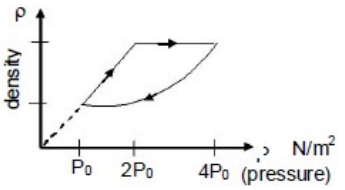
(A) The molar heat capacity of gas in the process $A \to B$ is $5\frac{R}{2}$ J/mol – K

(B) The molar heat capacity of gas in the process $A \to B$ is R J/mol – K

(C) The work done by the gas is $-400R$ J

(D) The work done by the gas is $-200R$ J.

7. An ideal gas undergoes cyclic process as shown in density pressure graph. During the process AB the work done $|W_{AB}| = 70J$. During the process BC, the gas absorbs 150 J of heat. During the process CA, gas undergoes expansion and does 210 J of work.



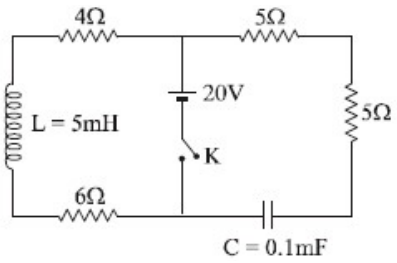
(A) The efficiency of cyclic process is 33%

(B) The efficiency of cyclic process is 66%

(C) The process CA is adiabatic.

(D) The process BC is isothermal.

8. In the circuit shown, the key (k) is closed at $t=0$, the current through the key at the instant $t = 10 - 3\ln 2 \sec$ is

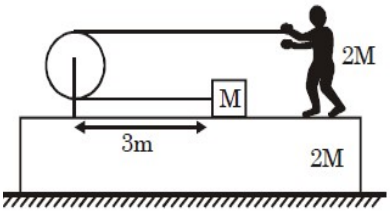


(A) $1.5\,\text{Amp}$
(B) $2.5\,\text{Amp}$
(C) $4\,\text{Amp}$
(D) $2/3\,\text{Amp}$

9. Which of the following statement is incorrect for a spherical body rolling without slipping on a rough horizontal ground at rest?
(A) The acceleration of a point in contact with ground is zero.
(B) The acceleration of COM of the sphere may or may not be zero.
(C) Friction force may or may not be zero.
(D) Work done by friction w.r.t. ground on the body is zero.

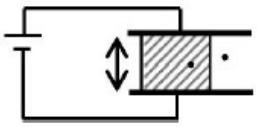
10. A smooth sphere A of mass m collides elastically with an identical sphere B at rest. The velocity of A before collision is 8 m/s in a direction making $60°$ with the line of centres at the time of impact.
(i) The sphere A comes to rest after collision.
(ii) The sphere B will move with a speed of 8 m/s after collision.
(iii) The directions of motion A and B after collision are at right angles.
(iv) The speed of B after collision is 4 m/s. The correct option is:-
(A) (i), (ii)
(B) (ii), (iii), (iv)
(C) (iii), (iv)
(D) (ii), (iii)

11. A block of mass M is tied to one end of massless rope. The other end of rope is in the hands of a man of mass 2M as show in figure. Initially the block and the man are resting on a rough plank of mass 2M as shown in figure. The whole system is resting on a smooth horizontal surface. The man pulls the rope. Pulley is massless and frictionless. What is the magnitude of displacement of the plank when the block meets the pulley (Man does not leave his position on the plank during the pull).



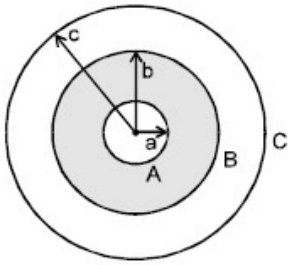
(A) 0.5m
(B) 0.6m
(C) 1.5m
(D) 2m

12. A parallel plate capacitor, partially filled with a dielectric slab of dielectric constant K, is connected with a cell of emf V volt, as shown in the figure. Separation between the plates is D. Then



(A) Electric field at point P is less than that at point P'
(B) Electric field at point P' is less than that at point P.
(C) Electric fields at points P and P' are equal
(D) Electric field at point P is $E=\dfrac{V}{KD}$

13. A and C are concentric conducting spherical shells of radius a and c respectively. A is surrounded by a concentric dielectric radius a, outer radius b and dielectric constant k. If sphere A be given a charges Q, the potential at the outer surface of the dielectric is :-



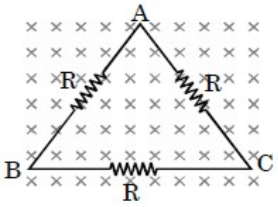
(A) $\frac{Q}{4\pi\varepsilon_0 kb}$

(B) $\frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{a}+\frac{1}{k(b-a)}\right)$

(C) $\frac{Q}{4\pi\varepsilon_0 b}$

(D) None of these

14. Three resistances of magnitude R each are connected in the form of an equilateral triangle of side a. The combination is placed in a magnetic field $B=B_0e^{-\lambda t}$ perpendicular to its plane. The induced current in the circuit is given by:



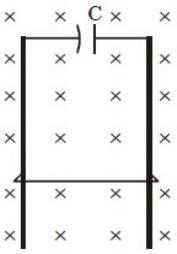
(A) $\left(\frac{a^2\lambda}{2\sqrt{3}R}B_0\right)e^{-\lambda t}$

(B) $\left(\frac{a^2\lambda}{4\sqrt{3}R}B_0\right)e^{-\lambda t}$

(C) $\left(\frac{a^2B_0}{\lambda 4\sqrt{3}R}\right)e^{-\lambda t}$

(D) $\left(\frac{a^2B_0R}{\lambda 4\sqrt{3}}\right)e^{-\lambda t}$

15. A wire of mass m and length ℓ can slide freely on a pair of smooth, vertical rails (figure). A magnetic field B exists in the region in the direction perpendicular to the plane of the rails. The rails are connected at the top end by a capacitor of capacitance C. The acceleration of the wire neglecting any electric resistance is:



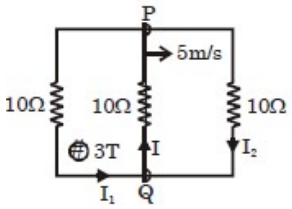
(A) $\frac{mg}{m+CB^2\ell^2}$

(B) $\frac{2mg}{m+CB^2\ell^2}$

(C) $\frac{mg}{CB^2\ell^2}$

(D) $\frac{mg}{2\left(m+CB^2\ell^2\right)}$

16. A rectangular loop has a sliding connector PQ length 2m and resistance 10Ω and it is moving with a speed 5 m/s as shown. The set-up is placed in a uniform magnetic field 3T going into the plane of the paper. The three currents I_1, I_2 and I are:



(A) $I_1=I_2=3\text{A}, I=1\text{A}$

(B) $I_1=I_2=5\text{A}, I=2\text{A}$

(C) $I_1=I_2=1\text{A}, I=2\text{A}$

(D) $I_1=I_2=1=2\text{A}$

17. A particle is projected with a velocity $(10\,\text{m/s})$ along y-axis from point (2, 3). Magnetic field of $\left(3\hat{i}+4\hat{j}\right)$. Tesla exist uniformly in the space. Its speed when particle passes through y-axis for the third time is: (neglect gravity)

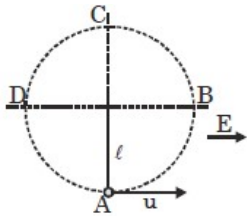
(A) $\sqrt{20}\,\text{m/s}$

(B) $\sqrt{60}\,\text{m/s}$

(C) $\sqrt{30}\,\text{m/s}$

(D) $10\,\text{m/s}$

18. A simple pendulum with a bob of mass $\text{m}=1\text{kg}$, charge $q=5\mu C$ and string length $\ell=1$ is given a horizontal velocity u in a uniform electric field $E=2\times10^6\ \text{V/m}$ at its bottom most point A, as shown in figure. It is given a speed u such that the particle leave the circular path at its topmost point C. Find the speed u. (Take g=10 m/s^2)



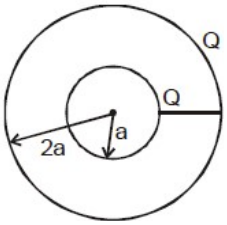
(A) $\sqrt{40}\,\text{m/s}$

(B) $\sqrt{50}\,\text{m/s}$

(C) $\sqrt{35}\,\text{m/s}$

(D) None of these

19. Figure shows a solid metal sphere of radius 'a' surrounded by a concentric thin metal hell of radius 2a. Initially both are having charges Q each. When the two are connected by a conducting wire as shown in the figure, then amount of heat produced in this process will be:



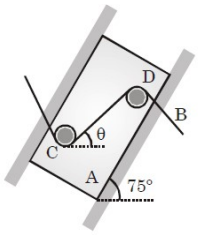
(A) $\dfrac{KQ^2}{2a}$

(B) $\dfrac{KQ^2}{4a}$

(C) $\dfrac{KQ^2}{6a}$

(D) $\dfrac{KQ^2}{8a}$

20. A slider block A moves downward at a speed of $v_A = 2\,m/s$, at an angle of $75°$ with horizontal as shown in the figure. The velocity with respect to A of the portion of belt B between ideal pulleys C and D is $v_{CD/A} = 2\,m/s$ at an angle θ with the horizontal. The magnitude of velocity of portion CD of the belt when $\theta = 15°$ is:



(A) $2\sqrt{3}\ m/s$

(B) $\sqrt{10}\ m/s$

(C) $2\sqrt{2}\ m/s$

(D) $2\ m/s$

SECTION – B
(Numerical Answer Type)

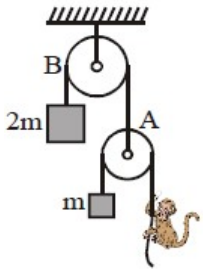
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. One mole of an ideal monoatomic gas is taken from state A to state B through a polytropic process in which gas obeys law $P = \frac{3}{2}T^{1/2}$. It is found that its temperature increases by 100 K in this process. Now it is taken from state B to C through another process for which internal energy is related to volume as $U = \frac{1}{2}V^{1/2}$, where volume at B is 100 m^3 and at C it is 1600 m^3. Find the total work performed by the gas (in Joule). Both the processes have proportionality constant unity with necessary dimensions. If your answer is x fill value of $\frac{x+45}{60}$. [Take: R = 8.3 J/mol-K]

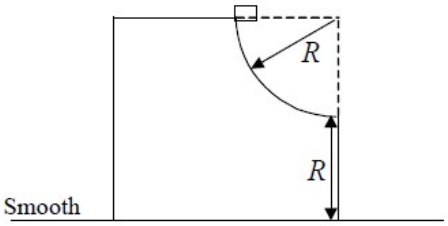
22. The reaction ${}^{7}_{3}Li + {}^{1}_{1}H \rightarrow {}^{7}_{4}Be + {}^{1}_{0}n$ is endothermic. Assuming that Li nuclei is free and at rest. What is the minimum kinetic energy (in keV) of incident proton so that this reaction occurs? Take Q value of this reaction as $-1645\,keV$. If your answer is N fill value $\frac{N}{235}$.

23. The end of a glass capillary tube of radius $r = 0.35\,mm$ lowered into the water to a depth of $h = 2.0\,cm$. What gauge pressure $\Delta p\,(\text{in Pa})$ is needed inside the capillary tube to blow a hemispherical air bubble at its lower end? Take surface tension of water as $7\times10^{-3}\ N/m$. Wetting is complete. Fill $\Delta p/40$ in OMR sheet.

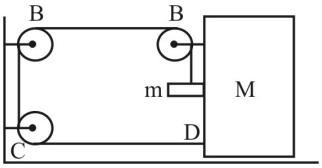
24. A monkey of mass m is balanced by a counterweight on the pulley A. Pulley A balances a load on the opposite side of pulley B (figure). The system is stationary. With what speed V (in m/s) will the block of mass $2m$ move. If the monkey starts climbing with a speed of $8\,m/s$ with respect to the rope? Rope and pulley are ideal.



25. A block is released from the state of rest from a wedge, as shown. The block and the wedge are both of the same mass. If friction is absent, the total horizontal displacement of the block with respect to earth before it falls on the earth is given by $0.5R+\sqrt{x}R$, then the value of x will be:



26. In figure the acceleration of m is $\frac{2\sqrt{x}mg}{M+5m}$, assuming that there is no friction between m and M, and all other surfaces are smooth and pulleys light. Find the value of x.

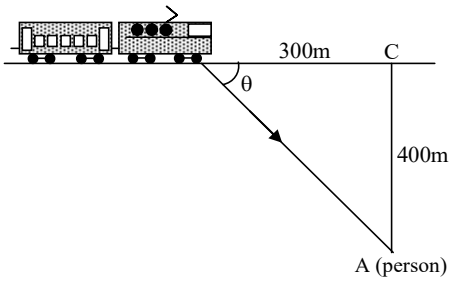


27. A non-conducting thin spherical shell of radius R has uniform surface charge density σ. The shell rotates about a diameter with constant angular velocity ω. The magnetic induction B at the centre of the shell is $\frac{2}{x}\mu_0\ \sigma\omega$ R. Find the value of x.

28. A gun kept on a straight horizontal road is used to hit a car travelling along the same road away from the gun with a uniform speed of 72 km h^{-1}. The car is at a distance of 500 m from the gun, when the gun is fired at an angle of 45° with the horizontal. Find the speed of projection of the shell from the gun. (in m/s^{-1})

29. An ideal monatomic gas is confined in a cylinder by a spring loaded piston of cross-section 8×10^{-3} m^2. Initially the gas is at 300K and occupies a volume of $2.4 \times 10^{-3} m^3$ and the spring is in its relaxed position. The gas is heated by a small electric heater until the piston moves out slowly by 0.1m. Calculate the heat supplied (in J) by the heater. The force constant of the spring is 8000 N/m and atmospheric pressure $1\times10^5\ \mathrm{N/m^2}$.

30. A train approaching a railway crossing at a speed of 120 km/h sounds a short whistle at frequency 640 Hz when it is 300m away from the crossing. The speed of sound in air is 340 m/s. What will be the frequency heard by a person standing on a road perpendicular to the track through the crossing at a distance of 400 m from the crossing. (in Hz)



Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which of the following equations does represent the velocity (v) of the ejected electrons when a metal is made to strike with light of frequency υ and threshold frequency of the metal is υ_0?
(m_e = mass of electron and h is Plank's constant)

(A) $\mathrm{v}=\sqrt{\dfrac{h(\upsilon-\upsilon_0)}{m_e}}$

(B) $\mathrm{v}=\sqrt{\dfrac{2h(\upsilon-\upsilon_0)}{m_e}}$

(C) $\mathrm{v}=\sqrt{\dfrac{h(\upsilon-\upsilon_0)}{2m_e}}$

(D) $\mathrm{v}=\sqrt{h(\upsilon-\upsilon_0)m_e}$

32. An element with mass number 181 contains 32% more neutrons as compared to protons. What is the symbol of that element?

(A) Pt
(B) Pd
(C) Au
(D) Hg

33. The electron gain enthalpy $\Delta_{\mathrm{eg}}\mathrm{H}$ of $\mathrm{Cl(g)}$ is $-349\ \mathrm{kJ\,mol^{-1}}$. If the ground state energy of $\mathrm{Cl(g)}$ is $\mathrm{x\ kJ\ mol^{-1}}$, the ground state energy (in kJ mol^{-1}) of $Cl^{-}(g)$ is:

(A) $x+349$
(B) x
(C) $x-349$
(D) $\dfrac{x-349}{17}$

34. Identify the correct set of molecules with different geometries and central atoms with different hybridisations:

(A) $SnCl_2, BeCl_2, OF_2$
(B) $H_2O, SO_2, HOCl$
(C) NH_3, H_2SO_3, XeO_3
(D) $SF_4,\ XeF_4,\ CF_4$

35. Observer the following molecules

$PCl_5, BrF_5, ClF_5, PF_5, ClF_3, XeF_4, XeF_2, IF_5$

The number of molecules having square pyramidal geometry from the above is:

(A) 4
(B) 5
(C) 3
(D) 6

36. If the kinetic energy and RMS speed of a gas at a certain temperature are $4.0\,kJ\,mol^{-1}$ and $5.0\times10^4\,cm\,s^{-1}$ respectively. The molecular weight of the gas is:

(A) 16
(B) 32
(C) 64
(D) 44

37. In how many of the following compound of sulphur, the oxidation state of sulphur atom is $+6$?

$H_2S_2O_8, H_2SO_5, H_2SO_3, H_2SO_4, H_2S_2O_7, SO_2Cl_2, SOCl_2$

(A) 3
(B) 5
(C) 4
(D) 6

38. Magnesium is burnt in air to form A and B. When B is hydrolyzed, C and D are formed D is the reactant in the manufacture of nitric acid by Ostwald's process. What is C?

(A) NH_3
(B) $Mg(OH)_2$
(C) MgO
(D) NO

39. Which of the following reactions can be used to prepare diborane?

(I) $BF_3 + LiAlH_4 \xrightarrow{\text{ether}}$
(II) $BF_3 + NaH \xrightarrow{450\,K}$
(III) $Na_2B_4O_7 + H_2O \longrightarrow$
(IV) $NaBH_4 + I_2 \longrightarrow$

(A) I, II, III
(B) II, III only
(C) III, IV only
(D) I, II, IV

40. Identify the correct statements
(I) Germanium exists only in traces.
(II) The order of electronegativity of Si, Ge, Sn is $Sn > Ge > Si$
(III) All group 14 elements are available in solid state
(A) I, II only
(B) II, III only
(C) I, III only
(D) I, II, III

41. In the detection of nitrogen of an organic compound by Lassaigne's test, Prussian blue colour is obtained. This is due to the formation of which of the following complexes?
(A) $Fe_2\left[Fe(CN)_6\right]$
(B) $Fe_4\left[Fe(CN)_6\right]_3$
(C) $Fe_3\left[Fe(CN)_6\right]_4$
(D) $Na_4\left[Fe(CN)_6\right]$

42. Identify Z in the following sequence of reactions,

$$C_2H_4Br_2 \xrightarrow[(ii)\ NaNH_2]{(i)\text{alc. KOH}} X \xrightarrow[\text{Fe tube 873K}]{\text{Red hot}} Y \xrightarrow[\text{Anhye. } AlCl_3]{(CH_3CO)_2O/\Delta} Z$$

(A) Acetophenone
(B) Anisole
(C) Toluene
(D) Chlorobenzene

43. The vapour pressure of chloroform $(CHCl_3)$, dichloromethane (CH_2Cl_2) at $298K$ are $200mm\ Hg$ and $415mm\ Hg$ respectively. An ideal solution is prepared by mixing $59.75g$ of $CHCl_3$ and $21.25\,g$ of CH_2Cl_2, the mole fractions of chloroform and dichloromethane in vapour phase respectively are:
(A) $0.509, 0.491$
(B) $0.491, 0.509$
(C) $0.201, 0.799$
(D) $0.799, 0.201$

44. Match the following:

List – I [Reaction]		List – II [Catalyst]	
(A)	$N_2(g)+3H_2(g)\longrightarrow 2NH_3(g)$	(I)	Ni
(B)	$2H_2(g)+O_2(g)\longrightarrow 2H_2O(l)$	(II)	Pt
(C)	$CO(g)+3H_2(g)\longrightarrow CH_4(g)+H_2O(g)$	(III)	$ZnO\text{-}Cr_2O_3$
(D)	$CO(g)+2H_2(g)\longrightarrow CH_3OH(g)$	(IV)	Fe

(A) A-III, B-II, C-I, D-IV
(B) A-IV, B-III, C-II, D-I
(C) A-IV, B-II, C-I, D-III
(D) A-IV, B-I, C-III, D-II

45. The pair of metals refined by “vapour phase refining” is:
(A) Ni, Cu
(B) Sn, Ni
(C) Zr, Ni
(D) Cu, Zr

46. White phosphorus, when heated with conc. $NaOH$ solution in an inert atmosphere of CO_2, form phosphine and a sodium salt of oxoacid of phosphorus, $'X'$. The oxidation state of phosphorus in $'X'$ is:
(A) $+3$
(B) $+4$
(C) $+1$
(D) $+5$

47. Sodium nitrite is reacted with H_2SO_4 to form $NaHSO_4.HNO_3$, water and X. Gold is dissolved in aqua regia to form water, $AuCl_4^-$ and $Y.X$ and Y are respectively.
(A) NO, NO_2
(B) NO_2, NO
(C) NO, NO
(D) N_2O, NO

48. Which of the following complex ions is most stable?
(A) $\left[Co\left(H_2O\right)_6\right]^{3+}$
(B) $\left[Co\left(CN\right)_6\right]^{3-}$
(C) $\left[Co\left(NH_3\right)_6\right]^{3+}$
(D) $\left[CoF_6\right]^{3-}$

49. The pair which does not exist is:
(A) (Tl II) TlI_2 and CuF_2
(B) (Tl I) TlI_3 and $CuBr_2$
(C) (Tl III) TlI_3 and CuI_2
(D) (Tl III) TlI_3 and $CuCl_2$

50. Match the following:

List – I		List – II	
(A)	Addition polymer	(I)	Bakelite
(B)	Condensation polymer	(II)	2-methyl-1, 3-butadiene
(C)	Acrilan	(III)	2, 3-dimethyl-1, 3-butadiene
(D)	Rubber	(IV)	Vinyl cyanide
		(V)	Polythene

(A) A-V, B-I, C-IV, D-II

(B) A-V, B-I, C-II, D-III

(C) A-I, B-V, C-IV, D-II

(D) A-I, B-V, C-II, D-III

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. Total number of different types of carbonyl compounds formed (Z) during ozonolysis of mixture of all the benzene derivative of formula C_8H_{10}.
[Present your answer X + Y + Z in the OMR sheet.]

52. Number of reaction which gives ester as major product.

(i) $R\text{-}CH{=}C{=}O \xrightarrow[H^+]{ROH}$

(ii) $H_3C{-}\underset{\underset{O}{\|}}{C}{-}CH_3 \xrightarrow{\text{m. CPBA}}$

(iii) $H_3C{-}\underset{\underset{O}{\|}}{C}{-}OH \xrightarrow{P_2O_5/\Delta}$

(iv) $H_3C{-}\underset{\underset{O}{\|}}{C}{-}O{-}\underset{\underset{O}{\|}}{C}{-}CH_3 \xrightarrow{O^-Et}$

(v) $\text{Acetaldehyde} \xrightarrow{\text{Aluminium alkoxide}}$

(vi) $RCOOR \xrightarrow{MeOH/H^+(1eq)}$

(vii) $H_2C{=}CH_2 + CH_3COOH \xrightarrow{\text{Silicotungtanic Acid}}$

(viii) $Me{-}\underset{\underset{O}{\|}}{C}{-}OH \xrightarrow{CH_2N_2}$

53. Solid $NH_4HS(s)$ (ammonium hydrogen sulphate) dissociates to give $NH_3(g)$ and $H_2S(g)$ and is allowed to attain equilibrium at 100°C. If the value of K_p for its dissociation is found to be 0.34. Find the total pressure at equilibrium.

54. Propane reacts with chlorine in sunlight to give two products. 1-chloropropane is obtained in 44% yield and 2-chloropropane is obtained in 56% yield of the total product. 2-Methylpropane reacts with chlorine under same conditions to produce 1-chloro-2-methylpropane 66% and 2-chloro-2-methylpropane 33%. What will be the percent yield (X) of the major product obtained when 1, 3, 5-trimethylcyclohexane is treated with Cl_2 in similar conditions?

55. Calculate the $|\text{lattice enthalpy}|$ in kJ mol^{-1} of $CaCl_2$, given that the enthalpy of:

(i) Sublimation of Ca is 121 kJ mol^{-1};

(ii) Dissociation of Cl_2 to Cl is 242.8 kJ mol^{-1};

(iii) Ionization of Ca to Ca^{2+} is 2422 kJ mol^{-1};

(iv) Electron gain enthalpy for Cl to Cl^- is -355 kJ mol^{-1};

(v) $\Delta_f H$ overall is -795 kJ mol^{-1}

56. NH_3 is heated at 15 atm from 25°C to 347°C assuming volume constant. The new pressure becomes 50 atm at equilibrium of the reaction $2NH_3 \rightleftharpoons N_2 + 3H_2$. Calculate % of moles of NH_3 actually decomposed.

57. Calculate $|\Delta_r G|$ in kJ mol^{-1} at 298K for the following reaction if the reaction mixture consists of 1 atm of N_2, 3 atm of H_2 and 1 atm of NH_3.

$N_2(g) + 3H_2(g) W \rightleftharpoons 2NH_3(g); \Delta_r G = -33.32\,kJ$

58. 0.71g of a sample of bleaching powder $(CaOCl_2)$ is dissolved in 100 mL of water. 50 mL of this solution is titrated with KI solution. The I_2 so liberated required 10 mL $Na_2S_2O_3$ (hypo) solution in acidic medium for complete neutralisation. Calculate the percentage of available Cl_2 from the sample of bleaching powder.

59. Pure benzene boils at 80°C. The boiling point of a solution containing 1g of substance dissolved in 83.4g of benzene is 80.175°C. If latent heat of vaporization of benzene is 90 cal per g, calculate the molecular weight of solute:

60. E.M.F. diagram for some ions is given ass:

$$FeO_4^{2-} \xrightarrow{E°=+2.20V} Fe^{3+} \xrightarrow{E°=+0.77V} Fe^{2+} \xrightarrow{E°=-0.445V} Fe^0$$

Determine the value of $E°_{FeO_4^{2-}/Fe^{2+}}$.

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The area value of a for which the integral $\int_{a-1}^{a+1} e^{-(x-1)^2}dx$ attains its maximum value is equal to:

(A) 1
(B) 2
(C) 3
(D) 4

62. Let $f(x)=\dfrac{e^{\ln x}.\operatorname{sgn}\left(x^2+x+1\right)}{\ln x}$. If $f(x)=k$ has one distinct solution then range of k is:

(A) $(-\infty,0)$
(B) $(0,e]$
(C) $(-\infty,0)\cup\{e\}$
(D) $[e,\infty)$

63. The value of $\sum_{n=1}^{\infty}\dfrac{3n^2+3n+1}{\left(n^2+n\right)^3}$ equal to:

(A) $\dfrac{1}{4}$
(B) $\dfrac{1}{8}$
(C) $\dfrac{1}{2}$
(D) 1

64. If $f(x)=\sin x+|\sin x|$ and $g(x)=f(x-\pi)+f(x+2\pi)$, then the value of $\int_{-2\pi}^{5\pi} g(x)dx$ is:

(A) 7
(B) 14
(C) 21
(D) 28

65. Let $f(x)=x^3+ax^2+ax+1$ has local extreme at $x=\alpha,\ \beta$ where $\alpha<\beta$. If $f(\alpha)+f(\beta)=2$, then the value of 'a' equals:

(A) 9

(B) $\frac{9}{2}$

(C) 7

(D) $\frac{7}{2}$

66. The value of definite integral $I=\int_0^{\pi/2} x\left|\sin^2 x-\frac{1}{2}\right|dx,$ is:

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{8}$

67. Number of roots of the equation $x^2.e^{2-|x|}=1$ is:

(A) 2

(B) 4

(C) 6

(D) Zero

68. If number of arrangements of the letters of the word MONOTONICITY in which O's do not appear adjacently is $9!(k),$ then k equals:

(A) 12

(B) 15

(C) 10

(D) 8

69. Number of solutions of the equation $\tan 3x-\tan 2x-\tan x=0$ in $[0,2\pi],$ is:

(A) 3

(B) 5

(C) 7

(D) More than 7

70. Coefficient of x^5 in the expansion of $\left(1+x+x^2\right)^{15}$ is:

(A) $3.\ {}^{16}C_4+{}^{16}C_5$

(B) ${}^{15}C_3+{}^{15}C_4$

(C) ${}^{15}C_5$

(D) ${}^{16}C_5$

71. The value of $4\cos\frac{2\pi}{7}\cos\frac{\pi}{7}-2\cos\frac{2\pi}{7}$ is equal to:

(A) $\frac{1}{2}$

(B) 0

(C) 1

(D) $\frac{3}{2}$

72. The sum of first ten terms of an arithmetic progression is 80 and its third term is 7. If the ninth terms is a_9, then $\left[a_9\right]$ is (where [.] denote greatest integer function):

(A) 1

(B) 7

(C) 8

(D) 9

73. If $A^2=A, B^2=B$ and $AB=BA$, then the number of distinct real values that $\mathrm{Det}\left(A-B\right)$ can take:

(A) 2

(B) 3

(C) 4

(D) Infinitely many

74. If $z_1^2+\left|z_1\right|=z_2^2+\left|z_2\right|=z_3^2+\left|z_3\right|=0\left(z_1\neq z_2\neq z_3\right)$ and z_1, z_2 and z_3 are the roots of equation $ax^3+bx^2+x+c=0,$ then:

(A) $a=1$

(B) $b=-1$

(C) $a=-1$

(D) $b=1$

75. An unbiased cubic die marked with 1, 1, 2, 2, 2, 4 is rolled 2 times. The probability of getting a total score of 4 if it is known that obtained total is an even number is

(A) $\frac{9}{19}$

(B) $\frac{9}{20}$

(C) $\frac{7}{20}$

(D) $\frac{5}{19}$

76. Centre of locus of point of intersection of tangents to the parabola $y^2 = 4ax$, the angle between them being always 45° is
(A) (3a, 0)
(B) (–3a, 0)
(C) (3a, a)
(D) (a, 3a)

77. Number of solutions of $\sin^{-1}\sqrt{x^2+x+1}+\cos^{-1}\sqrt{x^2-1}=\frac{3\pi}{4}$ is
(A) 0
(B) 1
(C) 2
(D) 3

78. The chord of contact of tangents from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$. Then a, b, c are in
(A) A.P.
(B) G.P
(C) H.P.
(D) None of these

79. If the lines $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{\lambda}$ and $\frac{x-1}{\lambda}=\frac{y-4}{2}=\frac{z-5}{1}$ intersect at a point (α, β, γ), then
(A) $\lambda = -1$
(B) $\lambda = 2$
(C) $\lambda = -3$
(D) $\lambda = 1$

80. A tangent of slope 2 to the ellipse $\frac{x^2}{36}+\frac{y^2}{49}=1$ has an intercept on the y-axis of length:
(A) $-2\sqrt{58}$
(B) $\sqrt{193}$
(C) $2\sqrt{58}$
(D) None of these

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. Number of arrangement of a, b, c, d in which neither a, b nor c, d come together is _____

82. Let P(sin A, cot B) be the point on the unit circle centred at origin and tan C, cot C are real numbers where A, B, C $\in$ R and the minimum value of the expression $(\tan C - \sin A)^2 + (\cot C - \cot B)^2 = p - \sqrt{2}\,q$, then (p + q) is equal to _____

83. Let α be a complex number satisfying the equation $|z-2|=2$ and $\bar{z}-zi=2(1-i)$. Then the perimeter of triangle formed by $\alpha, \bar{\alpha}$ and 2 is $\left(i=\sqrt{-1}\ \&\ \text{take}\ \sqrt{2}=1.41\right)$

84. Let a,b,c,d be four positive integers in arithmetic progression such that $a<b<c<d$ and $ab=c+d-1$. The sum of all possible value(s) of 'a' is:

85. If the tangents drawn from a point on the $2x^2-y^2=2a^2-b^2$ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ make angle α and β with the positive x-axis then find $\tan\alpha.\tan\beta$

86. A nine digit number is formed from $1,2,3,4,5$ such that product of all digits is always 1920. The total number of ways is $393\left({}^{9}P_{r}\right)$, where the value of r is_____

87. The sum $\sum_{i=0}^{5m}\binom{20}{i}\binom{30}{5m-i}$, ${}^{p}C_{q}=\binom{p}{q}=0$ if $p<q$, is maximum, then m equals_____

88. Let $\mathrm{f}(x)$ be a function such that $f(x)+f(x+2)=\sqrt{3}f(x+1)$ for all $x\in R$, then f is periodic with period T. Then value of T is_____

89. If the sum of first n terms of the A. P. $85,90,95,....$ is equal to the sum of first 3n terms of the A. P. $9,11,13,...,$ then $n+10$ equals_____

90. Let f be a continuous and differentiable function in (a,b), if $f(x).f'(x)\geq x\sqrt{1-\left(f(x)\right)^4}$ and $\lim\limits_{x\to a^+}\left(f(x)\right)^2=1$ and $\lim\limits_{x\to b^-}\left(f(x)\right)^2=\frac{1}{2}$.
Then the minimum value of $\left[a^2-b^2\right]+80$ is $\left([.]\to\text{represent greatest integer function}\right)$: